



## Group 19 probability work:

---

# GROUP 19 MEMBERS

MUGUME BRIGHTON

NASUBUGA SWABAT

NAMURUGI PRECIOUS WINNE

KIZZA MATHIAS

---

[Document subtitle]



## GROUP 19 PROBABILITY ASSIGNMENT

$$1. P(D) = 0.01 \text{ (DISEASE)}$$

$$P(T+/D) = 0.9 \text{ (TRUE POSITIVE)}$$

$$P(T-/D^1) = 0.9 \text{ (NO DISEASE)}$$

$$P(T+/D^1) = 0.1 \text{ (FALSE POSITIVE)}$$

We first find the total probability of testing positive

$$P(T+) = [P(T+/D) \cdot P(D)] + [P(T+/D^1) \cdot P(D^1)]$$

$$= (0.9 \cdot 0.01) + (0.1 \cdot 0.99)$$

$$= 0.108$$

We then get the probability you have disease given you are positive

$$P(D/T+) = \frac{P(T+/D) \cdot P(D)}{P(T+)}$$

$$P(D/T+) = \frac{0.009}{0.108}$$

$$P(D/T+) = 0.0833$$

Therefore the condition probability that the person actually has the disease is 8.33%

2. since every carton is equally likely to be picked at any position in the sequence. the probability of picking the 6<sup>th</sup> carton when its sour becomes:

$$P(\text{sour}) = \frac{\text{number of sour carton}}{\text{total number of carton}}$$

$$= \frac{2}{10}$$

$$= \underline{\underline{0.2}}$$

3. the statement at least one in the question means union

$$\text{Therefore } p(PUS) = \frac{1}{3}$$

$$P(PnS^1) = \frac{1}{9}$$

To find  $p(S)$ :

$$p(PUS) = p(PnS^1) + p(S)$$

$$\frac{1}{3} = \frac{1}{9} + p(S)$$

$$p(S) = \frac{2}{9}$$

## GROUP 19 PROBABILITY ASSIGNMENT

4. so to get the total no of combinations that can be formed:

We the total birth = 5 and the possibilities = 2

$$\text{Sample space} = 2^5 = 32$$

$$\text{But given that 1 boy} = 32 - 1 = 31$$

$$\text{Sample space} = 31$$

Probability of at least a girl is given birth ,we also eliminate the BBBBB

$$P(\text{at least 1 Girl}) = 31 - 1 = 30$$

$$P(G/B) = \frac{30}{31}$$

5. number of total television sets = 240

$$\text{Defective television sets} = 15$$

This is a problem of selection without replacement

$$\text{The probability 1}^{\text{st}} \text{ defective} = \frac{15}{240}$$

$$\text{The probability of the 2}^{\text{nd}} \text{ defective} = \frac{14}{239}$$

$$\begin{aligned} \text{The probability of both defective} &= \frac{15}{240} * \frac{14}{239} \\ &= \underline{\underline{0.00366}} \end{aligned}$$

6. using the principle  $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Where:

$$n(A \cup B) = 500 \text{ (everyone likes at least one)}$$

$$n(A) = 382 \text{ (like ice cream)}$$

$$n(B) = 362 \text{ (like cake)}$$

$$n(A \cap B) = ? \text{ (like both)}$$

$$500 = 382 + 362 - n(A \cap B)$$

$$N(A \cap B) = 244$$

The people liking both are 244 people

7. for two balls to be picked without replacement

The total = 5+3+7 =15

$$P(\text{both green}) = \frac{5}{15} * \frac{4}{14} * \frac{20}{210}$$

$$P(\text{both black}) = \frac{3}{15} * \frac{2}{14} = \frac{6}{210}$$

$$P(\text{both red}) = \frac{7}{15} * \frac{6}{14} = \frac{42}{210}$$

$$\begin{aligned} \text{Total probability} &= \frac{20}{210} + \frac{6}{210} + \frac{42}{210} = \frac{68}{210} \\ &= \underline{\underline{0.3238}} \end{aligned}$$

8 . The sample space tossing a coin until u see a head is :

(H.TH,TTH.TTTH,TTTTH)

This shows that it is a discrete because the outcomes can be counted

9.this involves tossing a coin forever

With a sample space  $S = (x_1, x_2, x_3, \dots)$

$x_i \in \{H, T\}$

therefore this is not a discrete ,because the number infinite sequences is uncountable